FOR OFFICIAL USE

N5

National
Qualifications
SPECIMEN ONLY

Mark

S844/75/01

Applications of Mathematics
Paper 1 (Non-calculator)

Date — Not applicable

Duration — 50 minutes

S844750 1

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day

Month

Year

Scottish candidate number

Total marks — 35

Attempt ALL questions.

You must NOT use a calculator.

To earn full marks you must show your working in your answers.

State the units for your answer where appropriate.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

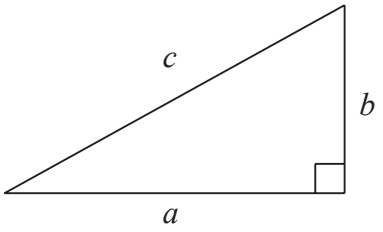
SQA ©

FORMULAE LIST

Circumference of a circle $C = \pi d$

Area of a circle $A = \pi r^2$

Theorem of Pythagoras



$a^2 + b^2 = c^2$

Volume of a cylinder $V = \pi r^2 h$

Volume of a prism $V = Ah$

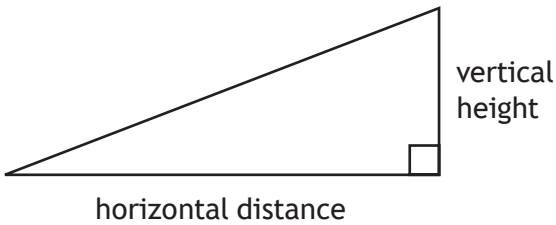
Volume of a cone $V = \frac{1}{3}\pi r^2 h$

Volume of a sphere $V = \frac{4}{3}\pi r^3$

Standard deviation $s = \sqrt{\dfrac{\Sigma(x-\bar{x})^2}{n-1}}$

or $s = \sqrt{\dfrac{\Sigma x^2 - \dfrac{(\Sigma x)^2}{n}}{n-1}}$, where n is the sample size.

Gradient



$$\text{gradient} = \frac{\text{vertical height}}{\text{horizontal distance}}$$

MARKS | DO NOT WRITE IN THIS MARGIN

Total marks — 35

Attempt ALL questions

1. Helen makes and sells candles.

These candles should be 22.5 cm tall.

She rejects any candle that is outwith the range of ±2 mm of this height.

Below are the heights, in centimetres, of 10 candles chosen at random.

22.2, 22.6, 22.5, 22.9, 22.3, 21.6, 22.6, 22.4, 22.7, 22.8

Calculate the percentage of candles that she rejects. **3**

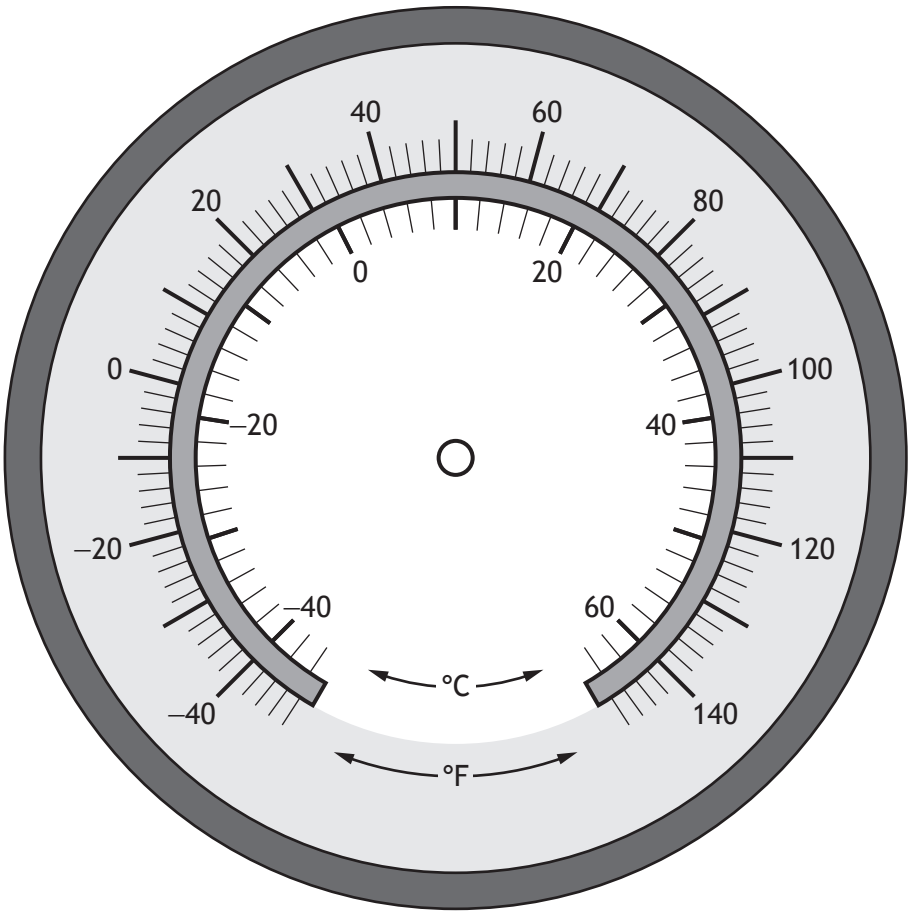
MARKS | DO NOT WRITE IN THIS MARGIN

2. Lynn is flying an aircraft and has been told that the outside temperature is 34 °C lower than the ground temperature.

The ground temperature is 6 °C.

Calculate the outside temperature and mark it on the gauge below. **2**

(An additional gauge, if required, can be found on *page 15*.)



MARKS | DO NOT WRITE IN THIS MARGIN

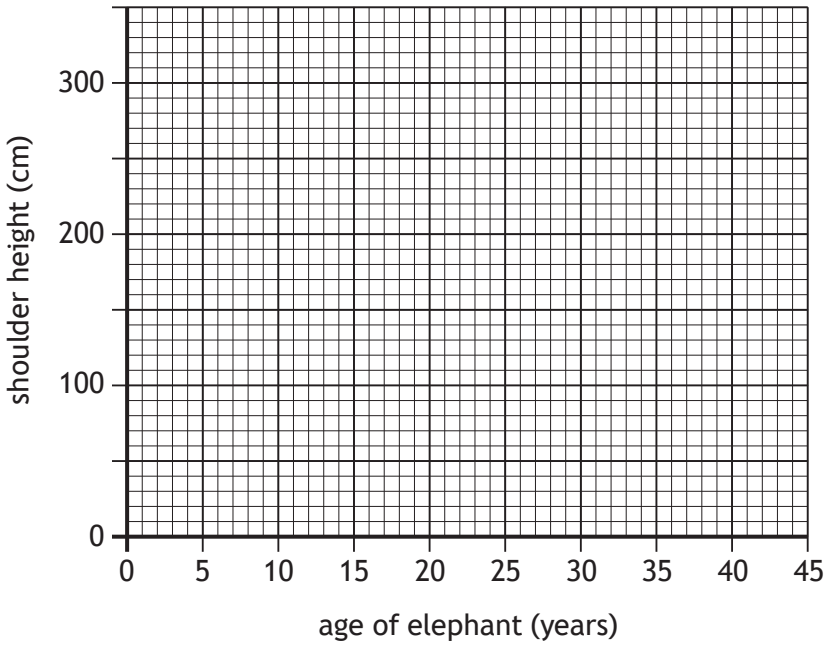
3. African elephants continue to grow for the duration of their lives.

The table below shows the age of a sample of African elephants and their shoulder heights.

Age of elephant (years)	12	17	28	35	43
Shoulder height (cm)	230	250	270	275	300

(a) On the grid below draw a scattergraph to show this data. **2**

(An additional grid, if required, can be found on *page 16*.)



(b) Draw a line of best fit on your scattergraph. **1**

(c) Use your line of best fit to estimate the age of an African elephant that has a shoulder height of 260 cm. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

4. Bryan, Jamie and Jessica bought **two** medium pizzas between them.

- Bryan ate $\frac{5}{7}$ of a pizza.
- Jamie ate $\frac{2}{3}$ of a pizza.
- Jessica ate the rest.

Calculate the total amount of pizza that Jessica ate. **3**

Give your answer as a fraction of a pizza.

MARKS | DO NOT WRITE IN THIS MARGIN

5. Steven flew to Hong Kong to start a new job.

The flight included a stop in Doha.

He flew from Edinburgh to Doha then from Doha to Hong Kong.

- The flight from Edinburgh to Doha took 6 hours 35 minutes.
- The flight from Doha to Hong Kong took 7 hours 20 minutes.
- Hong Kong is 8 hours ahead of Edinburgh.

Steven's plane took off from Edinburgh at 9:15 am local time.

It landed in Hong Kong at 8:50 am local time.

How long was the stop in Doha? **3**

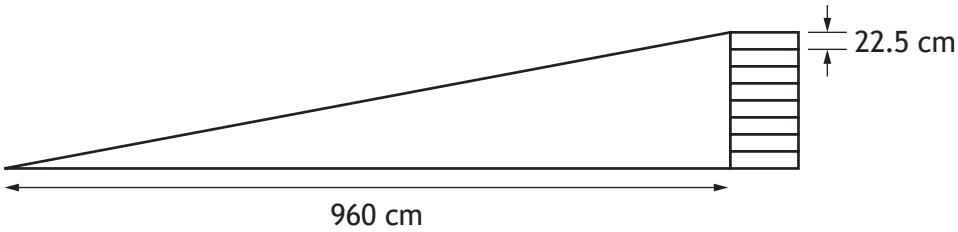
MARKS | DO NOT WRITE IN THIS MARGIN

6. John has a slope in his back garden.

The slope is the height of 8 planks.

Each plank is 22.5 cm in height.

The planks are 960 cm away from the bottom of the slope.



(a) Calculate the gradient of the slope. **2**

His neighbour Helen also has a slope.

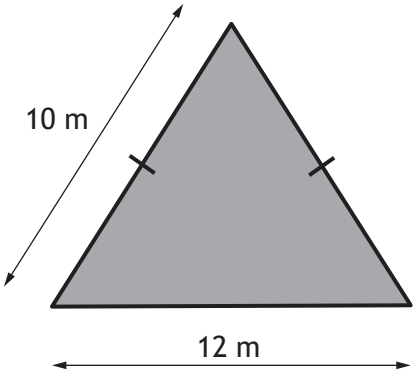
The gradient of her slope is 20%.

Helen thinks her slope is steeper than John's slope.

(b) Determine if she is correct. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

7. A lawn is to be created in the shape of an isosceles triangle with dimensions as shown below.



Calculate the area of the lawn. **3**

MARKS | DO NOT WRITE IN THIS MARGIN

8. Jack is going to a festival in the Czech Republic from his home in Glasgow.
His mum orders the tickets costing 1500 Czech koruna.
His mum lives in Poland so he must pay her back in Polish zloty.

Rates of exchange	
Pounds sterling (£)	**Other currencies**
1	30.00 Czech koruna
1	4.96 Polish zloty

Calculate how many Polish zloty he must give to his mum. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

9. Paul usually works 30 hours each week.

He is paid time and a half for any **additional** hours that he works.

His basic rate of pay is £12.50.

Last week, he worked a total of 37 hours.

(a) Calculate his gross pay for last week. **3**

MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

Paul is buying a new TV.

It is advertised at a price of £825.

He decides to use a payment plan to buy the TV.

The total cost of the TV using the payment plan is £845.80.

The payments are calculated as follows:

- deposit of $\frac{1}{5}$ of **advertised** price
- 8 equal monthly instalments
- final payment of £100.

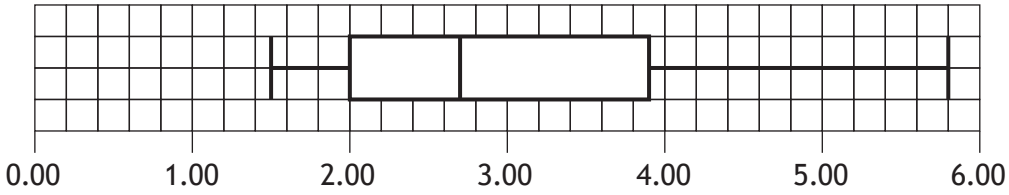
(b) Calculate the monthly instalment. **3**

MARKS | DO NOT WRITE IN THIS MARGIN

10. Mr Kenneth asked his class how much money they had spent on their lunch.

The results are shown in the boxplot.

amount of money spent on lunch (£)



(a) Calculate the inter-quartile range. **2**

The money spent on lunch by Mrs Campbell's class had an inter-quartile range of £1.82.

(b) Make one valid comment comparing the money spent on lunch by Mr Kenneth's class and Mrs Campbell's class. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

11. A company delivers parcels to people's homes.

The probability of a parcel arriving damaged is 0.023.

In one month, the company delivered 700 parcels, of which 15 were damaged.

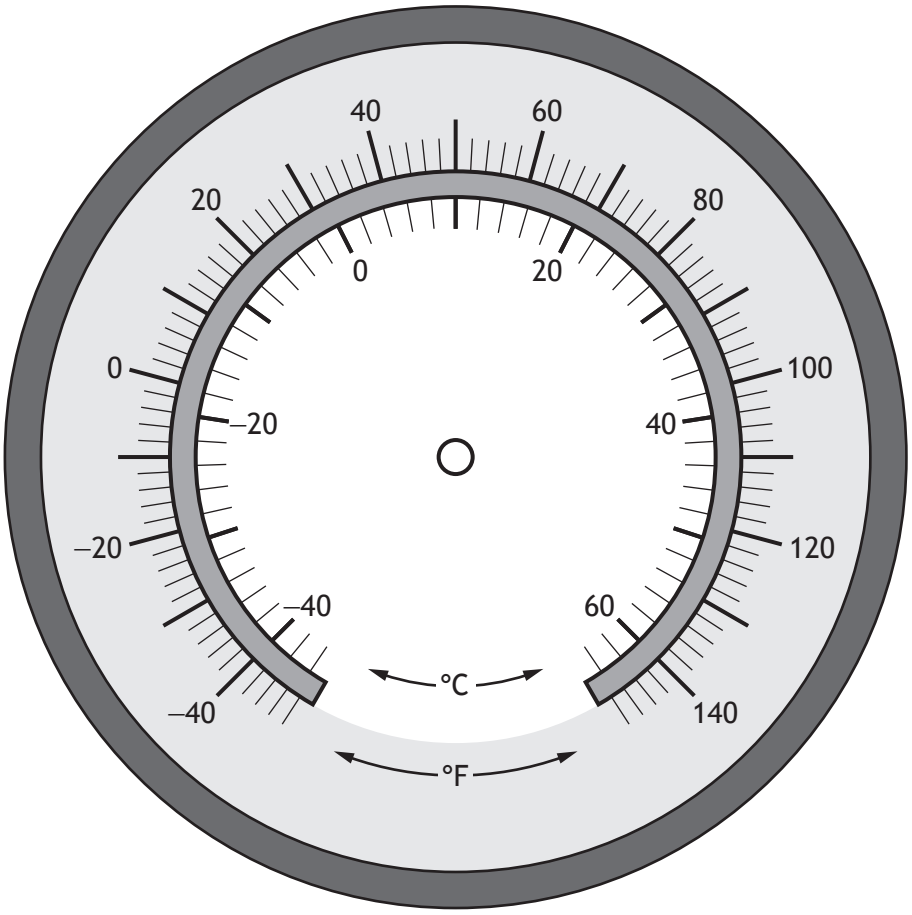
Determine if this is more or less than expected. **2**

[END OF SPECIMEN QUESTION PAPER]

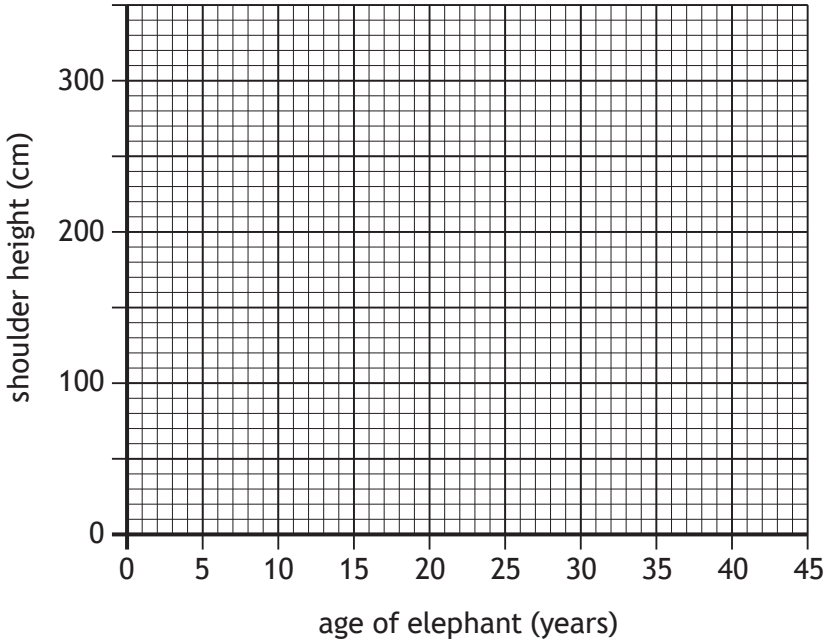
MARKS | DO NOT WRITE IN THIS MARGIN

ADDITIONAL SPACE FOR ANSWERS

Additional gauge for use with question 2



MARKS | DO NOT WRITE IN THIS MARGIN

ADDITIONAL SPACE FOR ANSWERS

Additional grid for use with question 3 (a)